

Treasure Hunt

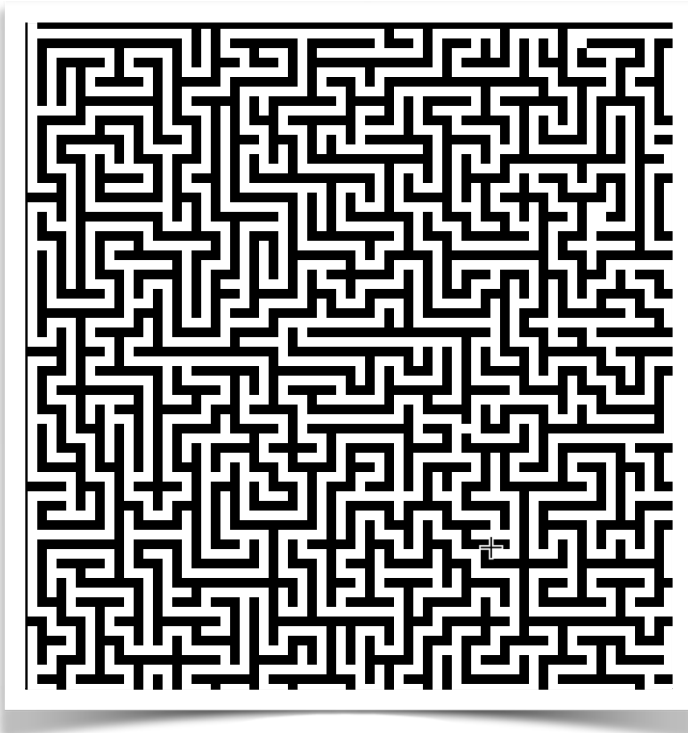
Each assignment is a reflection to one or more exercises from the past few weeks.
Read the assignments carefully and follow them to the letter.

Work alone. Do not borrow, generate or share code.

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1. Solve the maze...

If you zoom in enough, you see a maze like this. (Get your own from Blackboard)



From BB you can download a python file with 330 lines of code, called *maze.py*.

That is a script that opens the PNG-file with the maze, copies and saves the file under the name *solved-...png* and then tries to solve the maze.

The blue pixels are the pixels or nodes that have been visited by the AI to look for the right path. The red pixels show the shortest path found by the AI algorithm. The AI method to use is **Greed Best-First Search**.

Assignment 1: Write the code for the Greedy Best-First Search

Almost all of the work has been done for you in the given script, all you have to do is:

Fill in one single line of code in the **main()**, where you put the name of your PNG-file with the maze in it.

```
330
331 - if __name__ == "__main__":
332     # name to your image in same dir as this script
333     image_path = "xxx.png"
334     maze_runner(image_path)
335
```

Complete the **calc_manhattan()** method:

```
188     # finds the manhattan value of a given state
189     def calc_manhattan(self, state: tuple):
190         # TODO
191         # calculate the manhattan distance of this state to exit of the maze $
192
```

And program the part that does the actual AI searching, **GBfsFrontier()**:

```
275
276 - class GBfsFrontier:
277 -     def __init__(self, end: tuple):
278         # deque is a faster type of list for this context.
279         # https://dev.to/v_it_aly/python-deque-vs-listwh-25i9
280         # https://docs.python.org/3/library/collections.html#collections.deque
281         self.frontier = deque()
282
283     def add(self, node):
284         # TODO
285         # add the node somewhere to the frontier
286
287     def contains_state(self, state) -> bool:
288         # TODO
289         # check if the state is in the frontier, returns a boolean
290
291     def is_empty(self) -> bool:
292         # TODO
293         # checks if the frontier is empty, returns a boolean
294
295     def remove(self, current: Node|None=None, im: Imageclass|None=None):
296         # TODO
297         # searches somehow for the node to remove and to continue the search with
298
```

DO NOT CHANGE ANYTHING else in this Maze class.

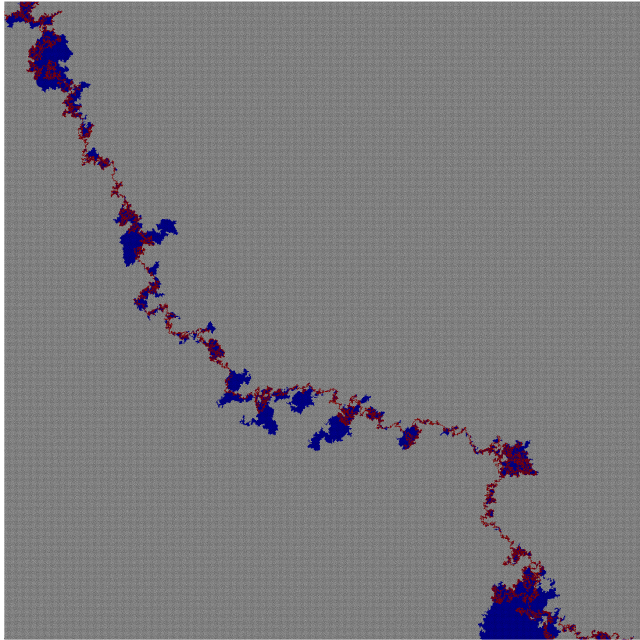
Before you start:

1. Before you start, you must install the **pillow** library in your IDE or virtual env.
 2. Change line 333 into the name of your image.
 3. Make sure that the image is in the same directory as the *maze.py* script.
-

Testing your results

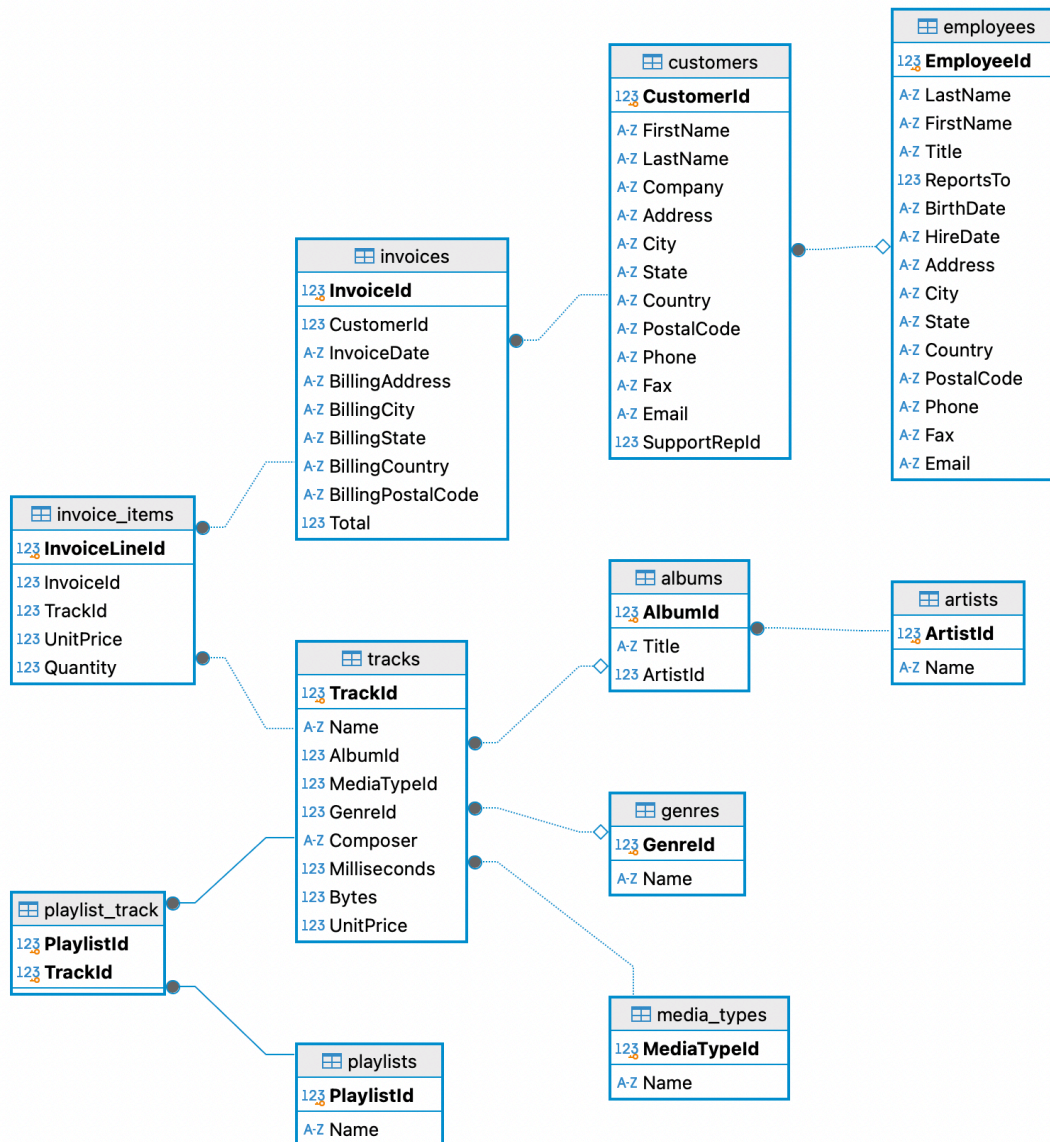
How can you tell if your solution is ok?

When solved, the file *solved-...png* shows the road from entrance to exit. Like in below image. Each student gets their own maze, so yours will look different than the one below.



2. Who's into Jazz?

For this SQL project you receive a SQLite database called *dbhunt.db*. The image *dbhunt.png* shows you the structure, the table names and the field names for all the tables.



You also receive a Python file called `sql_hunt.py`. This Python file does all the SQL work for you, all you have to do is come up with two SQL statements.

```
sql_hunt.py x
1  import sqlite3
2  import sys
3
4  dbname = "dbhunt.db"
5
6  connection = sqlite3.connect(dbname)
7  cursor = connection.cursor()
8
9  # ASSIGNMENT 1
10 # artists with Jazz tracks in the tracks table, order by artist name, a-z
11 # Output shows artists name and tracks name only
12 sql_statement_1 = """
13 select * from genres
14
15
16 """
17
18 # ASSIGNMENT 2
19 # the 5 artist with the most tracks in the tracks table of the Jazz genre, highest no. of tracks first.
20 # Output shows artists name and their no. of tracks only
21 sql_statement_2 = """
22
23
24 """
25
26 try:
27     results = cursor.execute(sql_statement_1) # and then 2 of course
28 except Exception as e:
29     sys.exit(e)
30 for r in results:
31     print(r)
```

Within the `"""` triple double quotes `"""`, where text is green in above image, you can type your SQL code, change the parameter in line 27 for the output of the first or second assignment.

The provided Python file as shown in above script will already give output.

Place in you virtual environment or VS Code

Upload or place all the files, including the `.db` file in your virtual environment or VS Code and run the code to test if it all works. You should get a lot of lines as output.

Assignment 2: Jazz artists

Generate a list of all the Jazz artists in the database, showing their names and all the tracks they have in the table with tracks of the genre Jazz.

Order this list by the names of the artists, a-z, where the artists name is mentioned first, then the track title.

Assignment 3: The top 5 jazz artists

Generate a list of the top 5 Jazz artists. That are the artists with the most tracks in the tracks table of the genre Jazz. Only the artists names should show, and the amount of Jazz tracks they have in the tracks table.

Testing your results

How can you tell if your solution is ok?

The results of Assignment 2 should look like the example below, but much longer.

```
('Billy Cobham', 'Solo-Panhandler')
('Billy Cobham', 'Do what cha wanna')
('Dennis Chambers', 'Roll Call')
('Dennis Chambers', 'Otay')
('Dennis Chambers', 'Groovus Interruptus')
('Dennis Chambers', 'Paris On Mine')
('Dennis Chambers', 'In Time')
('Dennis Chambers', 'Plan B')
('Dennis Chambers', 'Outbreak')
('Dennis Chambers', 'Baltimore, DC')
('Dennis Chambers', 'Talkin Loud and Saying Nothin')
('Gene Krupa', 'Jeepers Creepers')
('Gene Krupa', 'Blue Rythm Fantasy')
('Gene Krupa', 'Drum Boogie')
```

The results of Assignment 3 should look exactly like the example below:

```
('Miles Davis', 37)
('Gene Krupa', 22)
('Spyro Gyra', 21)
('Antônio Carlos Jobim', 14)
('Incognito', 13)
```

Upload to BB:

Upload the following files to BB:

1. The python script with name ***maze.py***.
2. A screen dump of the maze results with the name ***solved-yourname.png***.
3. The file ***sql_hunt.py*** containing your 2 SQL statements.
4. A screen dump **PNG** of the output of assignment 1, called ***assignment_2.png***.
5. A screen dump **PNG** of the output of assignment 2, called ***assignment_3.png***.

Do not put results in Word-files, PDF's, do not upload links to OneDrives or Google Drives, do not send emails or DM's or other creative solutions.

All uploads not according to above requirements are invalid.